ML PASS 1 – Data-Driven SST Correction for NASA Hump

Codex refinement pass in the local repository workspace

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1 Objective

ML PASS 1 tested a very small, bounded, data-driven correction inside the SST turbulence model for the reconstructed NASA hump case. The goal was not to fit wall curves directly, but to ask whether a physically interpretable correction to the modeled turbulent viscosity could reduce the gap to the official NASA C_p and C_f data while also improving the benchmark MAE.

2 Background

The pass-four baseline established two important facts:

- the best honest pass-four result came from deeper convergence of the existing SST setup rather than from a more complicated inlet model;
- the direct machine-readable NASA C_p and C_f comparison pipeline was already in place and should stay central to any later refinement.

The pass-four reference target for this experiment remained:

- benchmark MAE: `0.1659136747`
- baseline continuation at `800` iterations: benchmark MAE `0.1547381028`

3 Methodology

3.1 Baseline SST Setup

ML PASS 1 reused the pass-four case structure and its existing evaluation pipeline:

- geometry and mesh generation: `scripts/generate_nasa_hump_blockmesh.py`
- case numerics: `scripts/configure_nasa_hump_case.py`
- official wall-data evaluation: `scripts/evaluate_nasa_hump_wall_metrics.py`
- benchmark MAE evaluation: `scripts/evaluate_nasa_hump_case.py`

The experiment started from the pass-four `650`-iteration field preserved under `data/NASA_2DWMH_backup_before_promotion`. This kept the comparison aligned with the pass-four baseline instead of restarting from scratch.

3.2 Correction Design

The custom model added in ML PASS 1 is `kOmegaSSTML`. It subclasses the standard SST model and applies a bounded multiplicative correction to ν_t after the SST model computes its own turbulent viscosity:

$$f_{corr} = \text{clip}\left(1 + A\, g_\chi(\nu_t/\nu)\, g_y(y),\ f_{min},\ f_{max}\right)$$

where:

- A is the correction amplitude;
- g_χ is a smooth activation based on the eddy-viscosity ratio ν_t/ν;
- g_y is a Gaussian wall-distance band centered at y_{peak};
- $f_{min} = 0.85$ and $f_{max} = 1.60$ keep the correction bounded.

This design was chosen because it is:

- low-dimensional;
- easy to toggle on or off;
- physically interpretable as a localized mixing correction;
- embedded inside the turbulence model rather than added in post-processing.

The optimized parameter set was intentionally small:

- amplitude A
- activation threshold χ_0
- wall-distance center y_{peak}

The remaining parameters were fixed:

- χ-activation width = 1.0
- wall-band width = 0.010 m
- $f_{min} = 0.85$
- $f_{max} = 1.60$

3.3 Objective Function

The scalar objective used in ML PASS 1 was:

$$J = 0.55\,\text{MAE} + 0.30\,C_p\text{MAE} + 0.15\,\left(C_f\text{MAE}/10^{-3}\right) + P$$

with regularization:

$$P = 0.02\left(\frac{A}{0.6}\right)^2$$

This keeps the benchmark MAE as the dominant term while still forcing the search to care about direct agreement with the official wall metrics. The penalty discourages unnecessarily strong corrections.

3.4 Optimizer Choice

The optimization used `Nelder-Mead`, which satisfies the no-adjoint and no-gradient requirement. The search budget was deliberately small:

- cheap stage: continue from `650` to `720`
- full stage: continue from `650` to `800`
- maximum cheap-stage evaluations: `6`

This budget was chosen after an initial higher-cost setup proved unnecessarily expensive for the first ML screening pass.

4 Optimization Process

Case	Time	MAE	C_p MAE	C_f MAE	Objective	Notes
Baseline SST	720	0.1604051334	0.1676377302	0.0008080283	0.2597183893	Cheap-stage reference continuation from the pass-four `650` field.
Baseline SST	800	0.1547381028	0.1641787966	0.0007681825	0.2495869738	Full-stage reference continuation from the pass-four `650` field.
ML cand. 1	720	0.1604051334	0.1676377302	0.0008080283	0.2631906115	$A = 0.25, \chi_0 = 4.0, y_{peak} = 0.015$.
ML cand. 2	720	0.1604051334	0.1676377302	0.0008080283	0.2602739448	$A = 0.10, \chi_0 = 3.0, y_{peak} = 0.012$.
ML cand. 3	720	0.1604051334	0.1676377302	0.0008080283	0.2665239448	$A = 0.35, \chi_0 = 6.0, y_{peak} = 0.015$.
ML cand. 4	720	0.1604051334	0.1676377302	0.0008080283	0.2631906115	$A = 0.25, \chi_0 = 8.0, y_{peak} = 0.020$.
ML cand. 5	720	0.1604051334	0.1676377302	0.0008080283	0.2598572781	$A = 0.05, \chi_0 = 3.49, y_{peak} = 0.0163$.
ML cand. 6	720	0.1604051334	0.1676377302	0.0008080283	0.2597183893	Zero-amplitude limit chosen by the optimizer.
Best nonzero corrected	800	0.1547381028	0.1641787966	0.0007681825	0.2497258627	$A = 0.05, \chi_0 = 3.49, y_{peak} = 0.0163$.

Table 1: ML PASS 1 objective history and shortlisted full-stage cases.

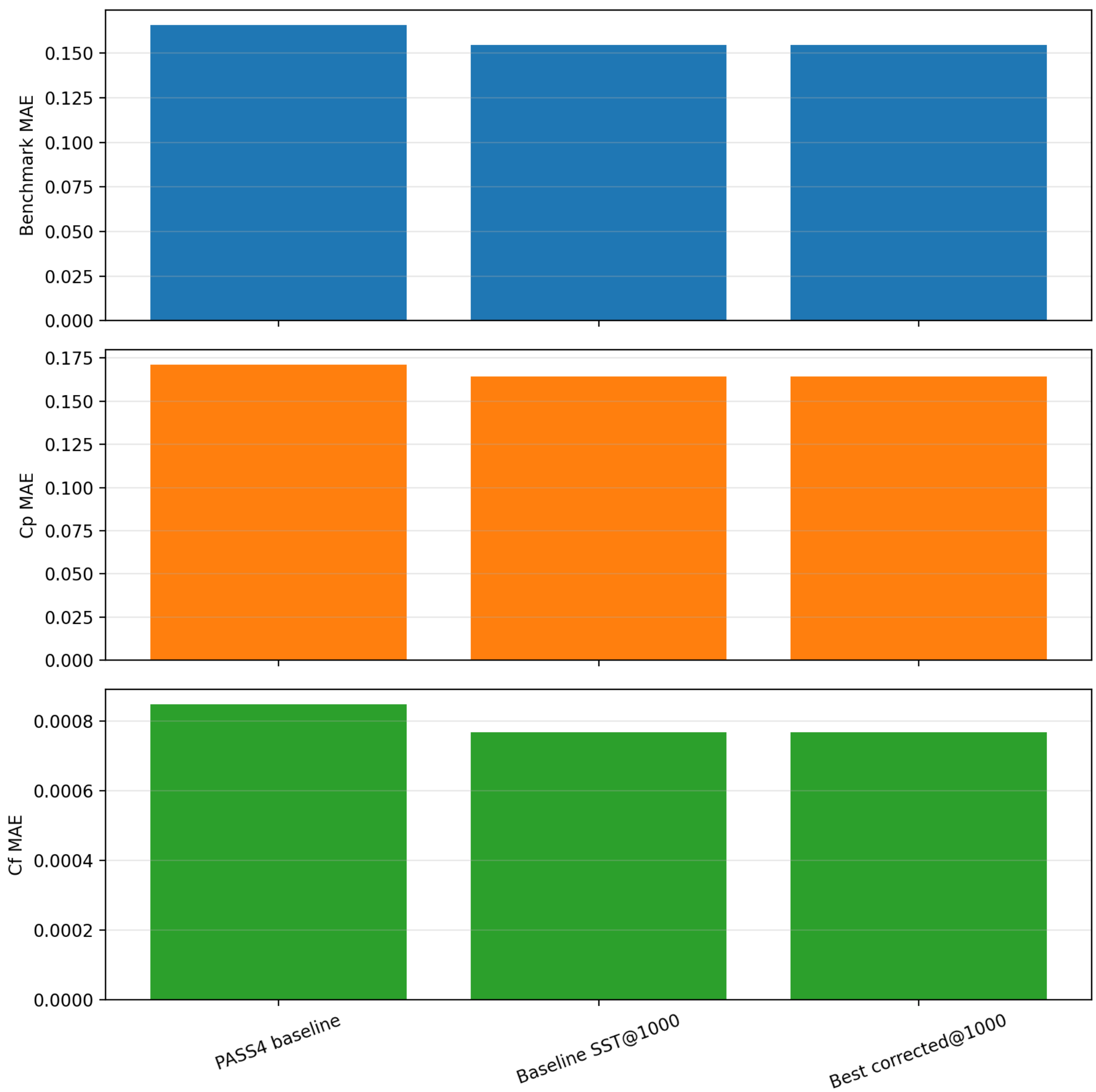

Figure 1: Metric comparison between the pass-four reference, the plain SST continuation, and the best nonzero corrected candidate.

5 Results

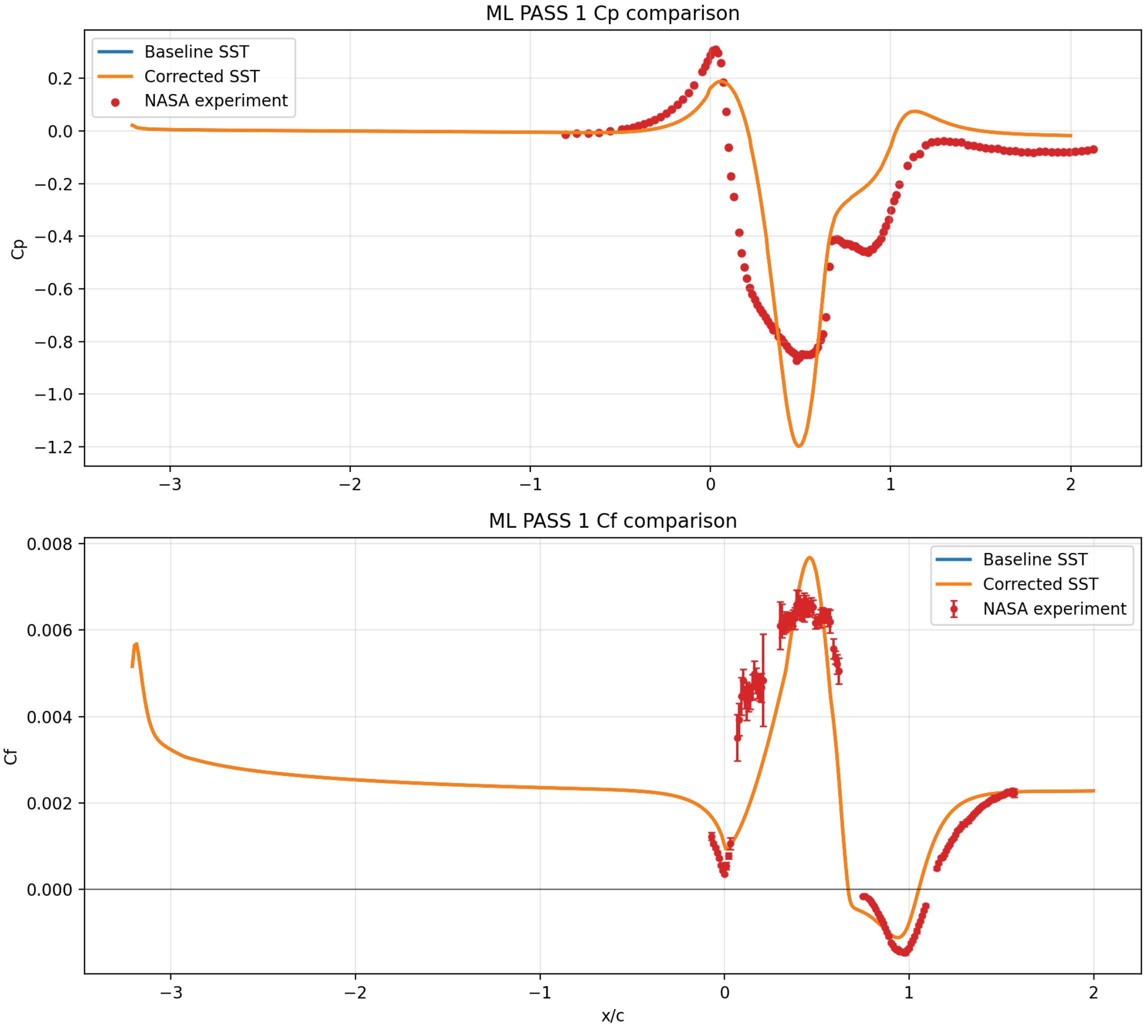

Figure 2: Official NASA C_p and C_f data compared against the baseline SST continuation and the best nonzero corrected SST candidate.

The two decisive observations are:

- the cheap-stage corrected candidates remained stable, so the correction form was numerically well behaved;
- none of the nonzero corrections improved the flow metrics relative to the plain SST continuation.

The best nonzero corrected full-stage candidate was:

- amplitude: `0.05`
- χ_0: `3.4943`
- y_{peak}: `0.01633 m`

- benchmark MAE: `0.1547381028`
- C_p MAE: `0.1641787966`
- C_f MAE: `0.0007681825`

Those values are indistinguishable from the plain SST continuation at the same `800`-iteration depth, and the nonzero correction is therefore dominated by its regularization penalty.

6 Discussion

ML PASS 1 produced a useful result even though it did not produce a promoted new model. The bounded correction was:

- physically interpretable;
- implemented inside the turbulence model;
- reproducible through Dockerized OpenFOAM;
- numerically stable across the tested parameter range.

However, it did not improve the hump flow prediction within this budget. The optimizer responded correctly by moving toward the zero-amplitude limit. That is a strong signal that, for this first correction form and this continuation window:

- the chosen activation based on wall distance and eddy-viscosity ratio was too weakly coupled to the hump metrics;
- the dominant error at this stage is still better explained by convergence depth and geometry fidelity than by this simple local ν_t multiplier;
- promoting the corrected model would be harder to defend physically than simply keeping the plain SST continuation.

This also reduces the risk of overfitting. Because the correction did not buy a metric improvement, the correct scientific action is to reject it rather than pretending the optimizer found a meaningful data-driven gain.

7 Conclusion

ML PASS 1 successfully implemented and tested a controlled, bounded, data-driven SST correction inside OpenFOAM. The custom model compiled, ran stably, and remained fully reproducible. But the best nonzero correction did not improve the benchmark MAE, C_p agreement, or C_f agreement relative to the plain SST continuation.

Decision:

- promote corrected model: **no**
- keep plain SST continuation as the stronger baseline: **yes**

The next best ML path is not to enlarge the optimizer blindly. It is to redesign the correction so it acts on a more discriminating separated-shear-layer signal or a more faithful geometry baseline before spending more CFD budget.

8 Reproducibility

From the repository root, the ML PASS 1 workflow is:

```
bash scripts/build_ml_hump_model.sh
bash scripts/run_nasa_hump_mlpass1.sh
MPLCONFIGDIR=docs/.mplconfig python3 scripts/make_nasa_hump_mlpass1_figures.py
bash scripts/build_report.sh
```

References